Models with Multiple Categorical Predictors (Factorial ANOVA).

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“Factorial” Designs

- In a fully factorial design, every level of one categorical variable (or factor) is combined with every level of another categorical variable (factor).

	A1	A2
B1	A1_B1	A2_B1
B2	A1_B2	A2_B2
B3	A1_B3	A2_B3

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B2	A1_B2	A2_B2
B3	A1_B3	A2_B3

Diet

Exercise	Control	Carb
Low	Low_Cont	Low_Carb
Med	Med_Cont	Med_Carb
High	High_Cont	High_Carb

Dealing with Multiple Factors

- This 3 (exercise) x 2 (diet) ANOVA design means that we effectively have six groups.
  - If we treated this as a one-way ANOVA, we would need 5 different codes:

	Low_Cont	Med_Cont	High_Cont	Low_Carb	Med_Carb	High_Carb
λ1	5	-1	-1	-1	-1	-1
λ2	0	4	-1	-1	-1	-1
λ3	0	0	3	-1	-1	-1
λ4	0	0	0	2	-1	-1
λ5	0	0	0	0	1	-1

Dealing with Multiple Factors

- Mathematically, this "works" because we have contrasts (the sum of each row is 0) and they are orthogonal (the sum all pairs of cross-products is zero):
  - $\sum \lambda_1 \lambda_2 = 5(0) - 1(4) - 1(-1) - 1(-1) - 1(-1) - 1(-1) = 0$
  - $\sum \lambda_4 \lambda_5 = 0(0) + 0(0) + 0(0) + 2(0) + 1(-1) - 1(-1) = 0$

	Low_Cont	Med_Cont	High_Cont	Low_Carb	Med_Carb	High_Carb
λ1	5	-1	-1	-1	-1	-1
λ2	0	4	-1	-1	-1	-1
λ3	0	0	3	-1	-1	-1
λ4	0	0	0	2	-1	-1
λ5	0	0	0	0	1	-1

Dealing with Multiple Factors

- However, a one-way ANOVA (treating all 6 groups as a single categorical factor) ignores families of related effects.
  - Is there an effect of diet, on average?

	Low_Cont	Med_Cont	High_Cont	Low_Carb	Med_Carb	High_Carb
λ1	5	-1	-1	-1	-1	-1
λ2	0	4	-1	-1	-1	-1
λ3	0	0	3	-1	-1	-1
λ4	0	0	0	2	-1	-1
λ5	0	0	0	0	1	-1

Dealing with Multiple Factors

- However, a one-way ANOVA (treating all 6 groups as a single categorical factor) ignores families of related effects.
  - Is there an effect of diet, on average?
  - Is there an effect of exercise, on average?

	Low_Cont	Med_Cont	High_Cont	Low_Carb	Med_Carb	High_Carb
λ1	5	-1	-1	-1	-1	-1
λ2	0	4	-1	-1	-1	-1
λ3	0	0	3	-1	-1	-1
λ4	0	0	0	2	-1	-1
λ5	0	0	0	0	1	-1

Dealing with Multiple Factors

- However, a one-way ANOVA (treating all 6 groups as a single categorical factor) ignores **families** of related effects.
  - Is there an effect of diet, on average?
  - Is there an effect of exercise, on average?
  - Does the effect of diet depend on exercise?

	Low_Cont	Med_Cont	High_Cont	Low_Carb	Med_Carb	High_Carb
λ1	5	-1	-1	-1	-1	-1
λ2	0	4	-1	-1	-1	-1
λ3	0	0	3	-1	-1	-1
λ4	0	0	0	2	-1	-1
λ5	0	0	0	0	1	-1

Dealing with Multiple Factors

- In theory, we could answer all of these questions through specific paired comparisons following the Omnibus F-test.
  - However, exhaustively comparing all of these groups would require **<u>21</u>** different comparisons!
  - $\alpha = 0.05 * 21 = 1.05$, a false positive is not just likely, its expected!

	Low_Cont	Med_Cont	High_Cont	Low_Carb	Med_Carb	High_Carb
λ1	5	-1	-1	-1	-1	-1
λ2	0	4	-1	-1	-1	-1
λ3	0	0	3	-1	-1	-1
λ4	0	0	0	2	-1	-1
λ5	0	0	0	0	1	-1

A Better Set of Contrasts

- Instead, we can use our contrasts more intelligently to create families of related tests.
  - Individual contrasts will answer our hypotheses more directly and the false positive rate for a given family of tests will be lower!

	Low_Cont	Med_Cont	High_Cont	Low_Carb	Med_Carb	High_Carb
λ1						
λ2						
λ3						
λ4						
λ5						

A Better Set of Contrasts

- **<u>The Main-Effect of Diet:</u>**
  - With only two levels, the **<u>main-effect</u>** of diet is the easiest to establish.
  - Codes needed: $k - 1 = 2 - 1 = 1$
    - $\lambda 1_k = (-1,1)$

	Low_Cont	Med_Cont	High_Cont	Low_Carb	Med_Carb	High_Carb
λ1	-1	-1	-1	1	1	1
λ2						
λ3						
λ4						
λ5						

A Better Set of Contrasts

- **<u>The Main-Effect of Exercise:</u>**
  - With three levels, the **<u>main-effect</u>** of exercise is trickier, but contrasts can be constructed identically to the one-way ANOVA contrast we have used before.
  - Codes needed: $k - 1 = 3 - 1 = 2$
    - $\lambda 2_k = (-1,0,1)$
    - $\lambda 3_k = (-1,2,-1)$

Together = Main Effect!

	Low_Cont	Med_Cont	High_Cont	Low_Carb	Med_Carb	High_Carb
λ1	-1	-1	-1	1	1	1
λ2	-1	0	1	-1	0	1
λ3	-1	2	-1	-1	2	-1
λ4						
λ5						

A Better Set of Contrasts

- **<u>The Interaction of Diet and Exercise:</u>**
  - To test the **<u>interaction</u>** we would multiply the interacting variables together the same way we did in continuous regression.
  - Codes needed: $k - 1 = 3 - 1 = 2$
    - $\lambda 4_k = \lambda 1_k \lambda 2_k$
    - $\lambda 5_k = \lambda 1_k \lambda 3_k$

Together = Interaction!

	Low_Cont	Med_Cont	High_Cont	Low_Carb	Med_Carb	High_Carb
λ1	-1	-1	-1	1	1	1
λ2	-1	0	1	-1	0	1
λ3	-1	2	-1	-1	2	-1
λ4	1	0	-1	-1	0	1
λ5	1	-2	1	-1	2	-1

A Better Set of Contrasts

- **<u>Orthogonal Contrast Codes for a fully Factorial Design:</u>**
  - Note that these codes are contrasts (because each row sums to zero) and they are orthogonal (i.e., uncorrelated) because the sum of the cross-products is zero.
    - $\sum \lambda_2 \lambda_3 = -1(-1) + 0(2) + 1(-1) - 1(-1) + 0(2) + 1(-1) = 0$
    - $\sum \lambda_4 \lambda_5 = 1(1) + 0(-2) - 1(1) - 1(-1) + 0(2) + 1(-1) = 0$

	Low_Cont	Med_Cont	High_Cont	Low_Carb	Med_Carb	High_Carb
λ1	-1	-1	-1	1	1	1
λ2	-1	0	1	-1	0	1
λ3	-1	2	-1	-1	2	-1
λ4	1	0	-1	-1	0	1
λ5	1	-2	1	-1	2	-1

Omnibus Tests versus Specific Contrasts

- **<u>Omnibus F for the full model:</u>** How much variance in the dependent variable was explained by all of our predictors as a set.

$$\boxed{Model\ A\text{:}\ y_i = \beta_0 + \beta_1(X_{1i}) + \beta_2(X_{2i}) + \beta_3(X_{3i}) + \beta_4(X_{4i}) + \beta_5(X_{5i}) + \epsilon_i}$$

$$\boxed{Model\ C\text{:}\ y_i = \beta_0 + 0(X_{1i}) + 0(X_{2i}) + 0(X_{3i}) + 0(X_{4i}) + 0(X_{5i}) + \epsilon_i}$$

Or… $\boxed{Model\ C\text{:}\ y_i = \beta_0 + \epsilon_i}$

Omnibus Tests versus Specific Contrasts

- **<u>Omnibus F for the Main-Effect of Diet:</u>** On average, was there an effect of diet?
  - Note that because Diet only had two levels (i.e., one contrast) the Omnibus F for this test is equivalent to testing the individual contrast.

$$Model\ A\text{:}\ y_i = \beta_0 + \beta_1(X_{1i}) + \beta_2(X_{2i}) + \beta_3(X_{3i}) + \beta_4(X_{4i}) + \beta_5(X_{5i}) + \epsilon_i$$

$$Model\ C\text{:}\ y_i = \beta_0 + \mathbf{0}(\boldsymbol{X_{1i}}) + \beta_2(X_{2i}) + \beta_3(X_{3i}) + \beta_4(X_{4i}) + \beta_5(X_{5i}) + \epsilon_i$$

Or… $$Model\ C\text{:}\ y_i = \beta_0 \qquad\qquad + \beta_2(X_{2i}) + \beta_3(X_{3i}) + \beta_4(X_{4i}) + \beta_5(X_{5i}) + \epsilon_i$$

- Thus, the numerator degrees of freedom (df1) for this test would be **<u>1</u>**.

Omnibus Tests versus Specific Contrasts

- **<u>Omnibus F for the Main-Effect of Exercise:</u>** On average, was there an effect of exercise?
  - Note that because exercise had three levels the Omnibus F for this test requires removing two contrasts.

$$Model\ A\text{:}\ y_i = \beta_0 + \beta_1(X_{1i}) + \beta_2(X_{2i}) + \beta_3(X_{3i}) + \beta_4(X_{4i}) + \beta_5(X_{5i}) + \epsilon_i$$

$$Model\ C\text{:}\ y_i = \beta_0 + \beta_1(X_{1i}) + \mathbf{0}(\boldsymbol{X_{2i}}) + \mathbf{0}(\boldsymbol{X_{3i}}) + \beta_4(X_{4i}) + \beta_5(X_{5i}) + \epsilon_i$$

Or…

$$Model\ C\text{:}\ y_i = \beta_0 + \beta_1(X_{1i}) \qquad\qquad\qquad\qquad + \beta_4(X_{4i}) + \beta_5(X_{5i}) + \epsilon_i$$

- Thus, the numerator degrees of freedom (df1) for this test would be **<u>2</u>**.

Omnibus Tests versus Specific Contrasts

- **<u>Omnibus F for the Interaction of Diet and Exercise:</u>** Did the effect of diet depend on exercise (or vice versa)?
  - Not that the interaction requires multiplying the interacting factors by each other. In this case, that takes 2 * 1 = 2 contrasts.

$$Model\ A\text{:}\ y_i = \beta_0 + \beta_1(X_{1i}) + \beta_2(X_{2i}) + \beta_3(X_{3i}) + \beta_4(X_{4i}) + \beta_5(X_{5i}) + \epsilon_i$$

$$Model\ C\text{:}\ y_i = \beta_0 + \beta_1(X_{1i}) + \beta_2(X_{2i}) + \beta_3(X_{3i}) + \mathbf{0}(\boldsymbol{X_{4i}}) + \mathbf{0}(\boldsymbol{X_{5i}}) + \epsilon_i$$

Or… $$Model\ C\text{:}\ y_i = \beta_0 + \beta_1(X_{1i}) + \beta_2(X_{2i}) + \beta_3(X_{3i}) + \epsilon_i$$

- Thus, the numerator degrees of freedom (df1) for this test would be **<u>2</u>**.